Ex no: 9
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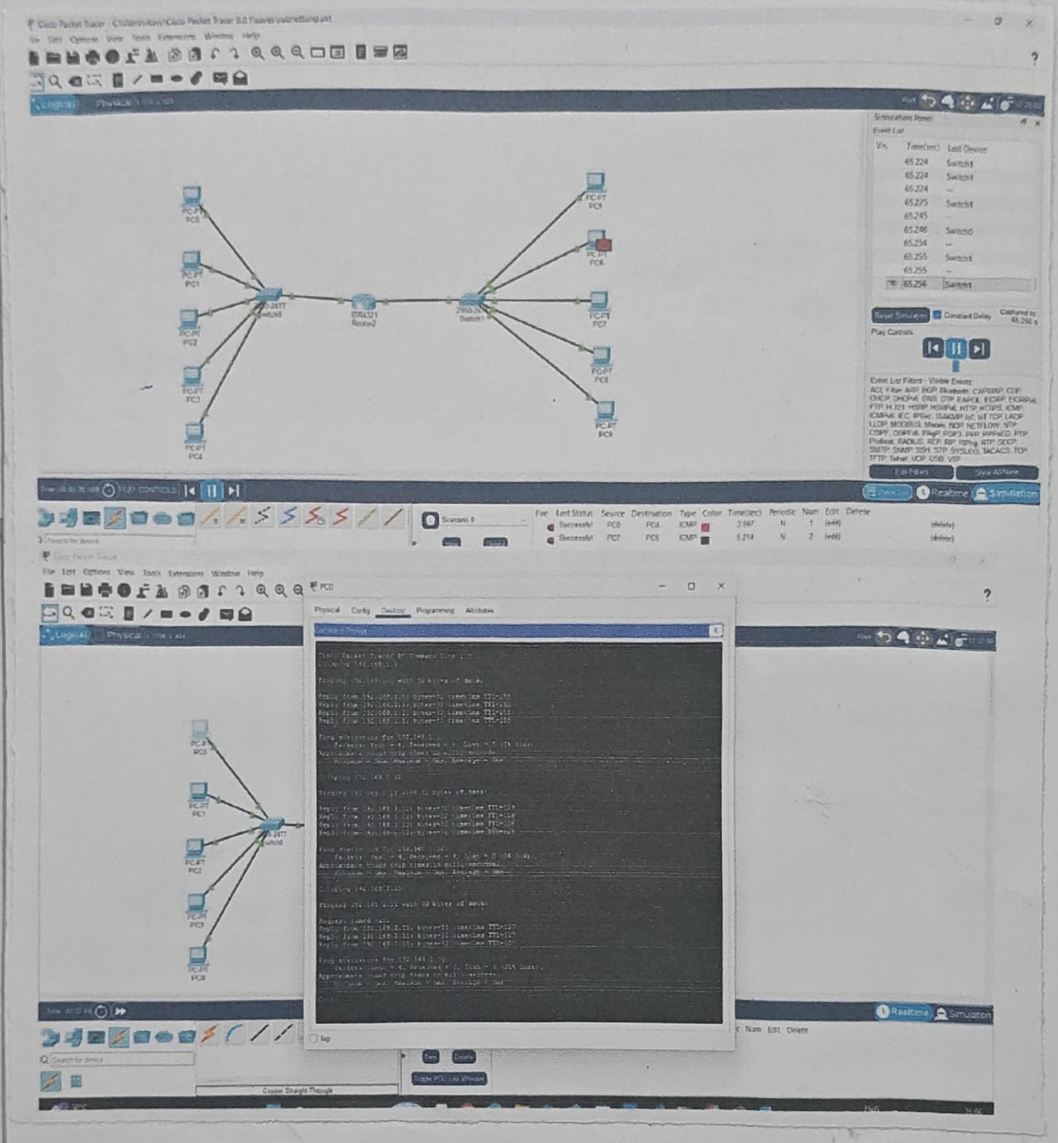
Implementation of Subnetting in Cisco Packet Tracer

Aim: To implement classless IP subnetting and configure a network using cisco packet tracer.

Subnetting divides a large IP network into smaller subnets using a suitable subnet mask.

Procedure:

1. Open Cisco packet Tracer and create a new network topology.
2. Add the required routers, switches and PCs.
3. Connect the devices using appropriate cables.
4. Divide the network into smaller /29 subnets.
5. Assign IP addresses, subnet masks and default gateways to the devices.
6. Configure the router interfaces using CLI
7. Configure switch ports in access mode
8. Configure each PC with its IP address, subnet mask.
9. Test Connectivity using ping command between PC's and routers.



Result: The different devices on a network were discovered successfully using Nmap in Tryhackme environment. Successfully implemented subnetting within a network and verified the network functioning.